

Geometry

Unit 12

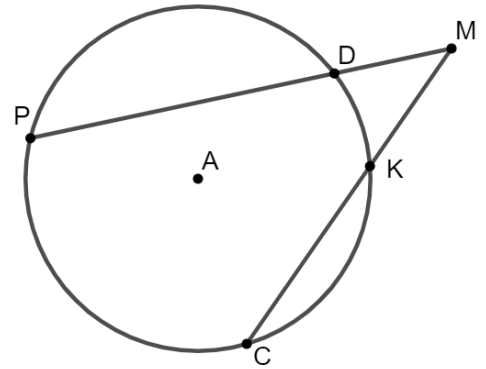
Lesson 2 Homework

Name Answer Key

Date _____

Period _____

Circle A with secants $\overline{PM}$ and $\overline{CM}$ is shown. $\overline{PM}$ and $\overline{CM}$ intersect at point M outside the circle so that $PD = x$, $DM = 3$, $CK = 8$, and $KM = 4$.



1. What is the length of secant $\overline{CM}$?

$$8 + 4 = CM = 12$$

2. Find the value of x .

$$3(x + 3) = 4(8 + 4)$$

$$3x + 9 = 48$$

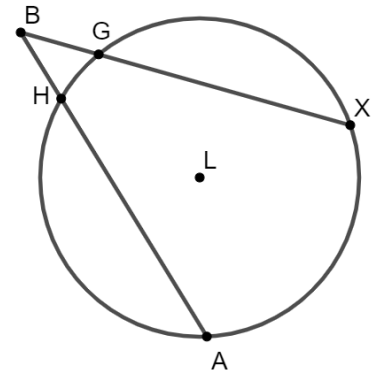
$$3x = 39$$

$$x = 13$$

3. What is the length of secant $\overline{PM}$?

$$(13) + 3 = PM = 16$$

Circle L is shown, where points G , X , A , and H lie on circle L . Secants $\overline{BX}$ and $\overline{BA}$ intersect at point B outside the circle such that $BG = 6$, $GX = 20$, $BH = 4$, and $HA = (6x - 7)$.



4. What is the value of x ?

$$6(20 + 6) = 4(6x - 7 + 4)$$

$$156 = 24x - 12$$

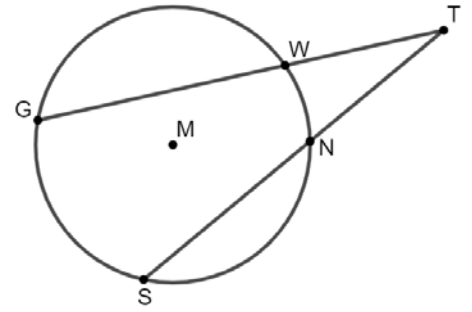
$$24x = 168$$

$$x = 7$$

5. Calculate the length of secant $\overline{BA}$.

$$4 + [6(7) - 7] = BA = 39$$

Circle M with secants $\overline{GT}$ and $\overline{ST}$ is shown. Secants $\overline{GT}$ and $\overline{ST}$ intersect outside the circle at point T so that $GW = 12x$, $WT = 4x$, $SN = 20$, and $NT = 16$.



6. Find the value of x .

$$4x(16x) = 16(36)$$

$$64x^2 = 576$$

$$x^2 = 9$$

$$x = 3$$

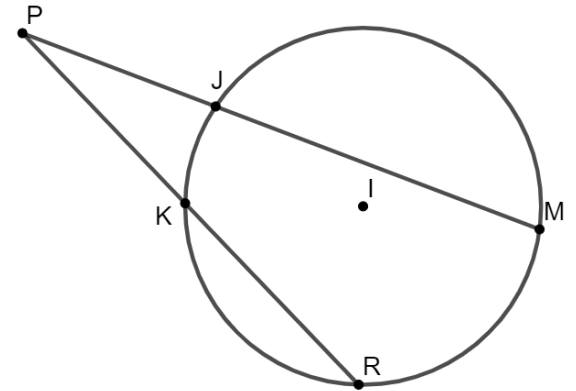
7. What is the length of secant $\overline{GT}$?

$$12(3) + 4(3) = \overline{GT} = 48$$

Carter and Isabella were solving for x in circle I where points J , M , R , and K are on the circle and secants $\overline{PM}$ and $\overline{PR}$ intersect at point P outside the circle. They know $PJ = 3$, $JM = (4x - 1)$, $PK = 4$, and $KR = 2x$.

Each of their equations are set up to solve for x .

Carter's Work	Isabella's Work
$3(4x - 1) = 4(2x)$	$3(4x + 2) = 4(2x + 4)$



8. Whose work is the correct equation needed to solve for x ?

Isabella's

9. Use this equation to find the value of x .

$$3(4x + 2) = 4(2x + 4)$$

$$12x + 6 = 8x + 16$$

$$4x = 10$$

$$x = 2.5$$

10. What is the length of secant $\overline{PR}$?

$$4 + 2(2.5) = \overline{PR} = 9$$